

Heart Failure Prediction Using Machine Learning

Project Overview: This project predicts heart failure using Machine Learning and Deep Learning models.

Objectives

- Analyze dataset
- Preprocess data
- Train Scikit-learn MLP and TensorFlow/Keras models
- Evaluate performance

Dataset

Clinical features including Age, Sex, Chest Pain Type, Blood Pressure, Cholesterol, ECG, Heart Rate and HeartDisease target.

Technologies

Python, Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow/Keras.

Workflow

Data Loading → Exploration → Visualization → Preprocessing → Model Training → Hyperparameter Tuning → Evaluation.

Evaluation Metrics

Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix.

Conclusion

The project demonstrates an end-to-end workflow for predicting heart disease risk using neural network models, helping support early diagnosis.